Harikrishnan Vinod

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LinkedIn | GitHub | Personal Website

WORK EXPERIENCE

Software Engineering Intern, Synapxe — Jan 2026-Jul 2026

- Contributed to the development and enhancement of national healthcare systems, including ACC and CCPS.
- Owned end-to-end delivery 6 features, translating requirements and Figma designs into React and TypeScript interfaces, building the .NET/C# backend APIs, integrating frontend and backend workflows, and testing each feature through completion.
- Diagnosed and resolved NEHR submission issues, reducing submission failures by 70%.
- Refactored data structures and field mappings across 8 FHIR resources, removing failure under UAT.
- Ensured accurate JSON-to-XML transformations for national healthcare integrations.
- Built and maintained integrations using React, TypeScript, .NET/C#, PostgreSQL, and Azure DevOps.

EDUCATION

Nanyang Technological University, Singapore — Aug 2023-May 2027

Bachelor of Computing (*Honours*), Computer Science with Minor in Business

- First-Class Honours (Expected) | GPA: 4.31/5
- Semester Exchange, Uppsala University, Sweden | Aug 2025 – Jan 2026
- Leadership: Hall 12 Socials Executive Committee; Faculty Sports Executive Committee
- Relevant Coursework: Data Structures and Algorithms, Software Engineering, Artificial Intelligence

Tampines Meridian Junior College — Jan 2019-Dec 2020

- 86.25 in A-Levels (2020) - Attained 4 Distinctions
- TM Gold Book Prize – Awarded for being top 5% in cohort for A-Levels

SKILLS

- Programming Languages: Python, Java, C, C++, C#, JavaScript, TypeScript, SQL
- Frontend: React, Next.js, HTML/CSS
- Backend & APIs: .NET, Node.js, Flask, REST APIs
- Databases: PostgreSQL, MySQL, Firebase Firestore
- Cloud & Tools: AWS, Azure, Google Cloud Platform (GCP), Docker, Azure DevOps, Git
- AI & Data: Artificial Intelligence, Machine Learning, Data Analytics
- Languages: Proficient in English, Hindi, Malayalam

PROJECTS & HACKATHONS

CycleGoWhere! — Jan 2025-May 2025

- Developed a web-based navigation app for cyclists designed to leverage Singapore's PCN and Smart Nation open datasets to give them safe, optimal cycling paths while accessing nearby amenities such as water coolers, repair shops, and bike parks within 500m.
- Engineered intuitive user features including ride activity logging, saved route management, and a dashboard for weekly and monthly cycling stats, enhancing user engagement and retention.
- Implemented real-time route planning and turn-by-turn navigation using OneMap API, integrated with interactive UI to deliver a seamless, responsive user interface
- Designed and deployed a modern full-stack application with React (Vite) and Flask, integrating Firebase Authentication and Firestore for secure, scalable user management and data storage.

HACKATHONS

NTU-WIT Hackathon Top 5 Feb 2025-Feb 2025

- Built an AI-assisted programming education platform that evaluates solutions based on their logic, allowing different valid implementations compared to strict character comparison checking enforced by traditional platforms.
- Developed the React and TypeScript frontend, Flask API, MySQL storage, and Perplexity-powered coding assistant.
- Placed Top 5 out of over 100 teams